

Cloud Kitchen Market Intelligence Project Methodology Report

1. Introduction

The objective of this project was to conduct a market intelligence study of restaurants operating in the College Road locality of Nashik. The project involved collecting restaurant data, investigating platform APIs, analysing menu information, cleaning the collected dataset, and generating business insights that could support the launch of a new cloud kitchen.

2. How Data Was Collected

Restaurant data was collected manually from Zomato by browsing restaurant listings and restaurant profile pages in the selected locality.

For each restaurant, the following information was collected:

- Restaurant Name
- Cuisine(s)
- Rating
- Number of Reviews
- Cost for Two
- Locality
- Delivery Time (if available)
- Restaurant Type

A total of 30 restaurant records were collected and stored in a CSV file named **raw_dataset.csv**.

For menu intelligence analysis, five restaurants were selected:

- Grill Craft Co.
- The Mykonos
- Hunger Wolf
- Café Nirvana
- Cream Stone

For each restaurant, the top 10 menu items, categories, and prices were recorded. Highest-priced items, lowest-priced items, and bestseller items were also identified.

3. Network Investigation

Chrome Developer Tools were used to inspect network requests generated while browsing restaurant pages on Zomato.

The Fetch/XHR filter was used to identify JSON-based API requests.

The following endpoints were documented:

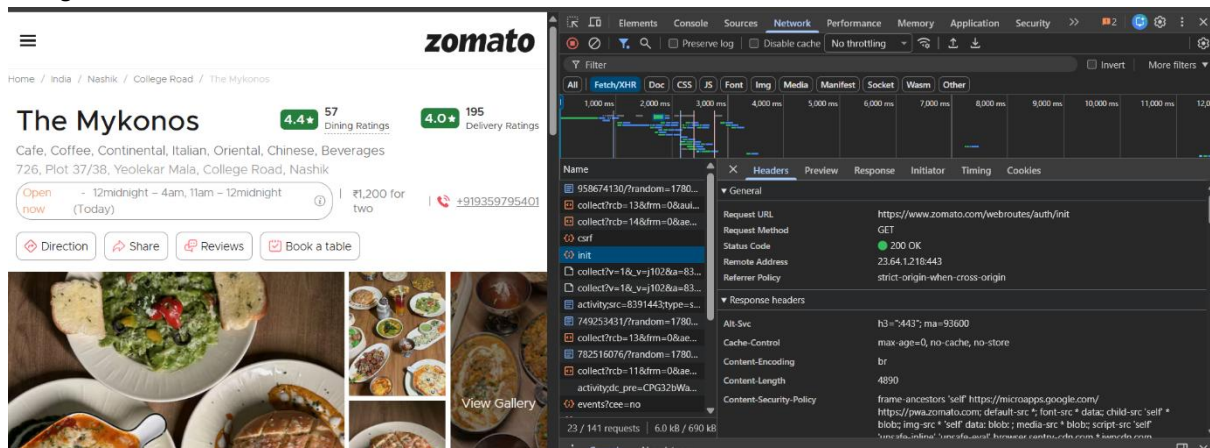
Endpoint 1

Request URL Pattern: /webroutes/auth/init

Method: GET

Response Type: JSON

Information Returned: Website initialization data, localization settings, currency information, and session configuration.



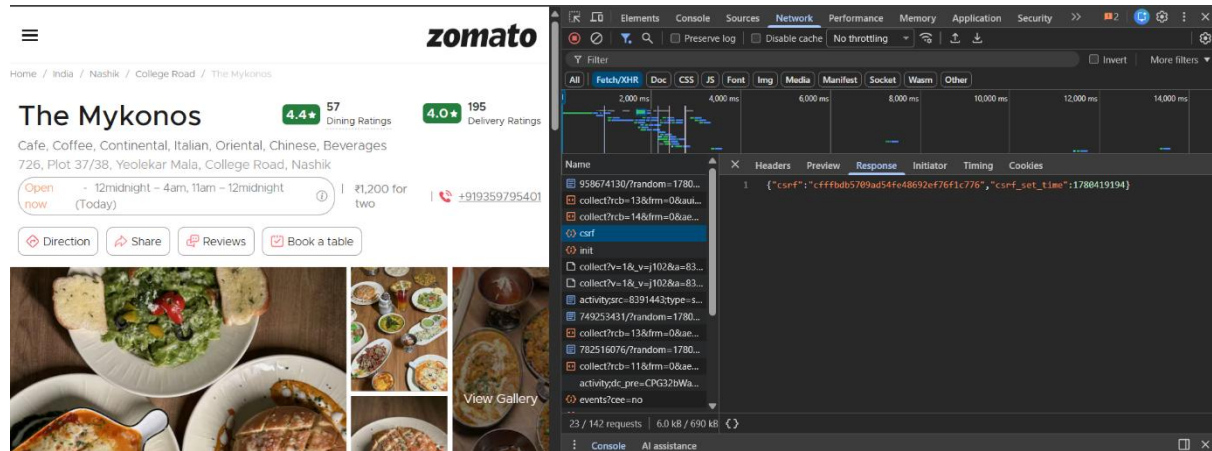
Endpoint 2

Request URL Pattern: /webroutes/auth/csrf

Method: GET

Response Type: JSON

Information Returned: CSRF token and session security information.



Endpoint 3

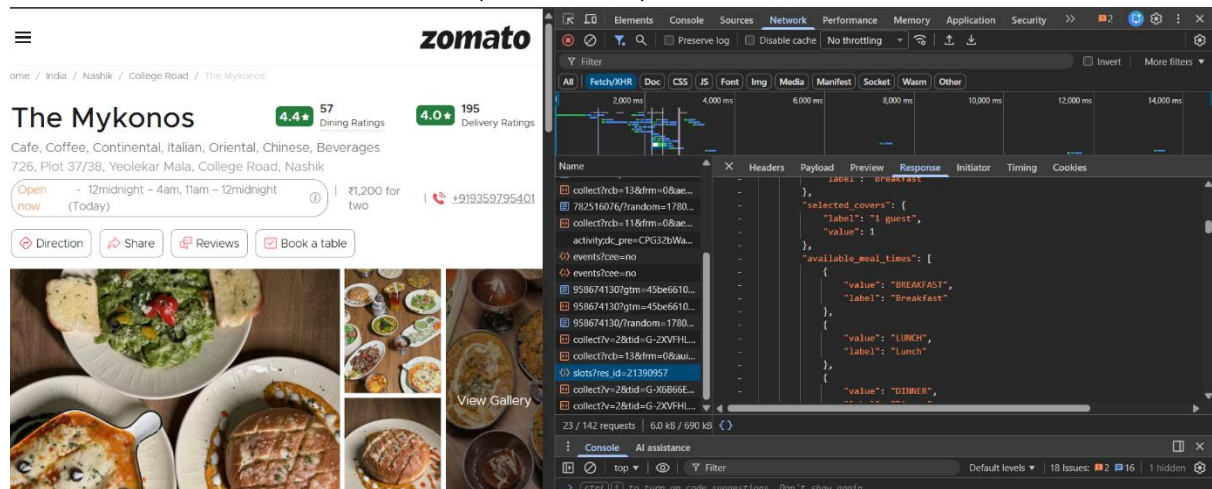
Request URL Pattern: /slots?res_id=...

Method: GET

Response Type: JSON

Information Returned: Restaurant reservation slots, meal timings, and booking availability information.

Screenshots of the Network Tab and JSON responses were captured as evidence.



4. Tools Used

The following tools were used during the project:

- Zomato
- Google Chrome
- Chrome Developer Tools (Network Tab)
- Microsoft Excel
- Visual Studio Code
- Google Colab
- SQL
- CSV Files
- PDF Documentation

5. Data Cleaning Process

The collected restaurant dataset was reviewed and cleaned to improve consistency and quality.

Before Cleaning

Total Records: 30

Cleaning Activities Performed

Duplicate Detection

Duplicate records were checked using:

- Restaurant Name
- Locality

No duplicate restaurant entries were found.

Missing Value Handling

The assessment required a Delivery Time field. However, during data collection, many restaurant profile pages on Zomato did not display delivery time information.

Since delivery times were not provided by the restaurants or visible on the source pages, no assumptions were made.

All unavailable delivery time values were replaced with:

"Not Available"

This ensured that the dataset remained complete while maintaining data accuracy.

Data Standardization

Restaurant type values were standardized to:

Dine-In Restaurant

Cuisine names were reviewed and normalized for consistency.

Examples:

- North Indian Cuisine → North Indian
- BBQ → Barbecue
- Cafe & Coffee → Cafe

After Cleaning

Total Records: 30

Records Removed: 0

The cleaned dataset was saved as **cleaned_dataset.csv**.

6. Problems Encountered

Several challenges were encountered during the project:

Missing Delivery Time Information

Many restaurant pages did not display delivery times. Since reliable delivery information was unavailable, these values were recorded as "Not Available" rather than estimated.

Bestseller Identification

Several restaurant menus did not explicitly display bestseller or most-ordered tags. In such cases, bestseller items were estimated using menu prominence and featured items.

API Investigation

Many network requests were related to analytics and tracking rather than restaurant data. Additional filtering and investigation were required to identify useful JSON endpoints.

Data Consistency

Some cuisines appeared under multiple naming conventions, requiring normalization during the cleaning process.

7. Assumptions Made

The following assumptions were made during the project:

1. Restaurants with physical locations and table-booking facilities were classified as Dine-In Restaurants.
2. Missing delivery time information was recorded as "Not Available".
3. Cost-for-two values displayed on Zomato were considered accurate and used without modification.
4. Bestseller items were estimated when explicit bestseller indicators were unavailable.
5. Restaurant information displayed on Zomato was assumed to be current and accurate.

8. Improvements With More Time

If additional time were available, the following improvements could be implemented:

- Increase the dataset size from 30 restaurants to over 100 restaurants.
- Include multiple localities for comparative analysis.
- Automate data collection using Python-based scraping techniques.
- Perform customer review sentiment analysis.
- Build interactive dashboards using Power BI or Tableau.
- Analyse pricing trends and cuisine popularity using statistical methods.
- Investigate additional APIs for deeper platform analysis.

9. Conclusion

This project provided practical experience in data collection, API investigation, menu intelligence, data cleaning, SQL schema design, and business analysis. The final dataset and findings provide useful insights into restaurant market conditions in the selected locality and can support decision-making for launching a new cloud kitchen business.